

Chapter

Sports Training

1. The process of preparation of a sportsman, based on scientific and technical principles, for higher performance is called : MAIN 2024

- (a) Sport times.
- (b) Sports Training.
- (c) Physical education.
- (d) Physical fitness.

2. The training by an athlete should be followed without any break. Which principle does it refer to? COMP 2021

- (a) Principle of Continuity
- (b) Principle of Adaptation
- (c) Principle of Overload
- (d) Principle of Progression

3. Which of the following exercises helps the body to gradually make the transition to a resting state? SQP 2023

- (a) Jogging
- (b) Cooling down
- (c) Warm-up
- (d) Sprinting

4. The principle of ____ refers to the process of the body getting used to training through regular repetition. MAIN 2020

- (a) Recovery
- (b) Adaptation
- (c) Reversibility
- (d) Specificity

5. In periodisation, the year is divided into how many periods? COMP 2022

- (a) Two
- (b) Four
- (c) Three
- (d) Six

6. What is the main purpose of conditioning training? MAIN 2021

- (a) To develop social skills
- (b) To improve physical strength, endurance, and flexibility
- (c) To replace technical training
- (d) To avoid injuries completely

7. Why is setting intermediate goals important in sports training? COMP 2023

- (a) To replace the need for professional coaching
- (b) To track progress in short-term and longterm training phases
- (c) To discourage athletes from making mistakes
- (d) To ensure that training remains easy and comfortable

8. What do you mean by Training? COMP 2024

9. What do you mean by Sports Training? SQP 2021

10. What is the main aim of Sports Training? MAIN 2023

11. What is Periodisation? SQP 2022

12. Assertion (A) : Conditioning training is essential for athletes as it improves strength, endurance, and flexibility.

Reason (R) : Conditioning training helps athletes develop sport-specific technical skills.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

13. Assertion (A) : The principle of overload states that training intensity should be increased beyond normal activity levels to improve performance.

Reason (R) : If training intensity remains constant without progressive overload, the athlete will continue to improve at the same rate. **COMP 2023**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

14. Assertion (A) : The principle of reversibility states that if training is stopped for an extended period, fitness levels will decline.

Reason (R) : Regular training ensures that fitness gains are maintained, and long breaks may result in loss of strength, endurance, and flexibility. **SQP 2021**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

15. Assertion (A) : Tactical training in sports focuses on improving endurance and strength.

Reason (R) : Tactical training is primarily concerned with strategies, game plans, and decision-making. **MAIN 2022**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is false, but R is true.
- (d) A is true, but R is false.

16. What does the principle of continuity emphasize? MAIN 2022

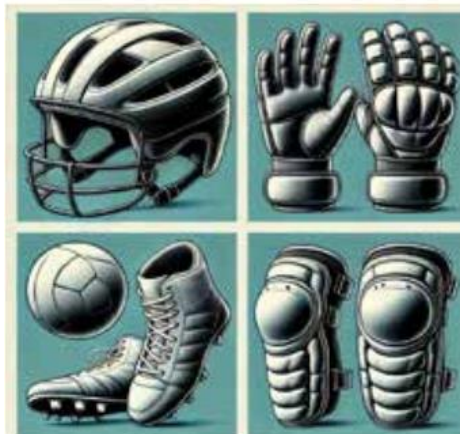
- (a) Training should include long breaks for rest
- (b) Training should be a continuous process with minimal gaps
- (c) Training should only be conducted in specific seasons
- (d) Training must always be high-intensity

17. Match the Following : Principles of Sports Training

Column I (Principles)		Column II (Description)	
(I)	Principle of Overload	1.	Variety helps prevent boredom and injuries.
(II)	Principle of Continuity	2.	Increasing training intensity gradually.
(III)	Principle of Variance	3.	Training must be specific to the sport.
(IV)	Principle of Specificity	4.	Training should be continuous.

- (a) I-1, II-2, III-3, IV-4
(b) I-4, II-3, III-2, IV-1
(c) I-1, II-3, III-4, IV-2
(d) I-2, II-4, III-1, IV-3 **COMP 2024**

18. Choose the Correct Protective Equipment for Injury Prevention Which of the following protective equipment is best suited for preventing head injuries in sports?



- (a) Shin guards
(b) Gloves
(c) Helmet
(d) Knee pads **SQP 2023**

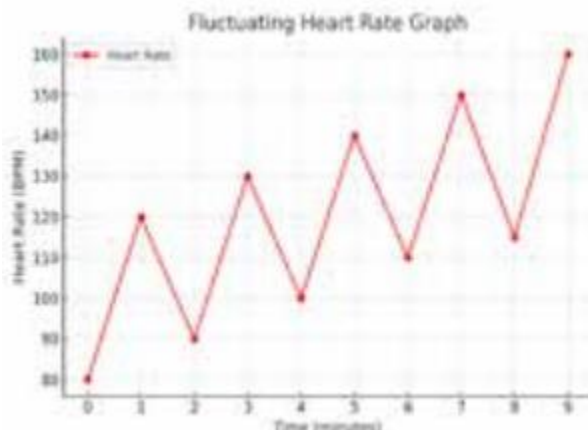
19. According to the Principle of ____, taking extended breaks from training can lead to a decline in fitness levels. MAIN 2022

- (a) Continuity
- (b) Overload
- (c) Recovery
- (d) Reversibility

20. The concept of periodisation divides training into structured cycles, typically lasting ____ weeks, followed by a recovery phase.

- (a) One
- (b) Three
- (c) Five
- (d) Seven

21. Identify the Training Method Based on the Heart Rate Graph A heart rate monitor graph shows a continuous high-intensity effort followed by brief rest periods. What type of training does this represent? COMP 2022



- (a) Continuous training
- (b) Interval training
- (c) Circuit training
- (d) Strength training

22. Match the Following : Factors Affecting Training Response

Column I (Factors)			Column II (Sequenced Order)
(I)	Genetics	1.	Environmental conditions like altitude, temperature.
(II)	Nutrition	2.	Affects energy levels and muscle repair.
(III)	Motivation	3.	Determines athletic potential and recovery rate.
(IV)	Environment	4.	Psychological factor affecting performance.

(a) I-1, II-2, III-3, IV-4

(b) I-3, II-2, III-4, IV-1

(c) I-1, II-4, III-2, IV-3

(d) I-2, II-3, III-1, IV-4 **SQP 2021**

23. State the objectives of training in sports.

COMP 2021

24. What are the important components for a successful training programme?

SQP 2023

25. Explain the principle of overloading and recovery in sports training.

COMP 2022

26. Why is year-round fitness important for an athlete?

MAIN 2021

27. What does the term 'Training' mean in Sports?

MAIN 2024

28. A football coach is designing a comprehensive training program for his team. The program includes strength training to enhance physical endurance, skill drills to refine techniques, agility exercises for better movement and mental conditioning to improve focus and reduce performance anxiety. He believes that a wellrounded approach will help the players perform better during matches. **MAIN 2024**

(i) What type of training is being implemented in this scenario?

(ii) Name and explain two key elements of this training program and their importance in football performance.

29. A professional sprinter wants to improve his race times and overall performance. His trainer develops a structured training plan that includes sprint drills for speed enhancement, resistance training for strength and flexibility exercises to reduce injury risks. The trainer ensures that the plan is sport-specific and aligns with the athlete's goals. **COMP 2021**

- (i) Which principle of sports training is being applied in this situation?**
- (ii) Why is it essential for sprinters to follow a training plan that aligns with their specific sport requirements?**

30. A basketball player has taken a long break due to an ankle injury. After resuming training, he notices a decline in his endurance, muscle strength, and reaction speed. His coach explains that a lack of continuous training can lead to a loss of fitness and skill efficiency over time. **SQP 2023**

- (i) Which principle of sports training does this scenario highlight?**
- (ii) Suggest two strategies that can help an athlete maintain fitness levels during periods of injury or prolonged inactivity.**

31. A cricket team is undergoing specialized training that includes batting drills, bowling techniques, and fielding exercises. The coach emphasizes that this training is tailored to improve the technical and tactical skills required in cricket, rather than focusing on general fitness alone. **COMP 2022**

- (i) Which type of sports training is emphasized in this scenario?**
- (ii) How does sport-specific training benefit athletes compared to general physical fitness training?**

Solution

1. (b) Sports Training

Sports training is the scientific and systematic process used to prepare athletes for higher performance in sports. It focuses on improving skills, endurance, strength, and overall performance through planned techniques and methods.

2. (a) Principle of Continuity

The **Principle of Continuity** states that training should be regular and without long breaks. Continuous practice helps maintain and improve fitness levels, prevents reversibility, and ensures steady progress in performance.

3. (b) Cooling down

Cooling down includes light exercises performed after intense activity. It helps the body slowly return to a resting state by reducing heart rate, preventing muscle stiffness, and improving recovery.

4. (c) Reversibility

Reversibility indicates that the body adjusts to training stimuli through repeated activities over time.

5. (c) Three Periodisation divides the year into three cycles to maximize training effectiveness and results.

. **Conditioning training** prepares the body for sports performance by enhancing key physical attributes.

- **Main purposes:**

1. **Strength:** Builds muscles to perform demanding tasks.
2. **Endurance:** Improves stamina to sustain activity for longer periods.

3. **Flexibility:** Increases the range of motion to prevent injuries.

- Examples include weight training, running, stretching, and plyometric exercises.

6. (b) To improve physical strength, endurance, and flexibility

Conditioning training aims to enhance an athlete's physical attributes, including muscular strength, cardiovascular endurance, and flexibility. This helps athletes perform better in their sport, reduces fatigue, and lowers the risk of injuries. It does not replace technical training or focus on social skills.

7. (b) To track progress in short-term and long-term training phases

Intermediate goals help athletes monitor their progress in smaller, manageable phases. They act as milestones that provide motivation, allow assessment of performance, and help make adjustments in training if needed. For example, a long-distance runner may first aim to improve their 5K time before attempting a marathon.

8. Training is the systematic process of preparing an individual for a specific task or activity. In sports, it involves exercises, drills, and techniques aimed at enhancing performance, skills, strength, and endurance.

9. Sports training is a planned and structured program designed to improve an athlete's physical, technical, psychological, and intellectual abilities. It prepares sports persons for higher performance through systematic practice and exercises.

10. The main aim of sports training is to help athletes achieve their highest performance potential by developing their physical, technical, intellectual, psychological, and moral abilities through systematic exercises and techniques.

11. Periodisation is the systematic planning of athletic or physical training, involving the progressive cycling of training aspects over a specific period to achieve peak performance in the most important competition of the year.

12. (c) A is true, but R is false.

Conditioning training focuses on enhancing an athlete's physical attributes like strength, endurance, and flexibility. Development of sport-specific technical skills occurs through separate technical training, not conditioning.

13. (c) A is true, but R is false.

The principle of overload emphasizes gradually increasing training intensity to stimulate improvement. If intensity remains constant, the athlete will not continue to improve; progress will plateau.

14. (a) Both A and R are true, and R is the correct explanation of A.

The principle of reversibility states that stopping training for a long period leads to a decline in fitness. Regular training maintains strength, endurance, and flexibility, preventing loss of performance.

15. (c) A is false, but R is true.

Tactical training focuses on strategies, game plans, and decision-making in sports. It does not aim to improve endurance or strength, which are developed through conditioning training.

16. (b) Training should be a continuous process with minimal gaps

The principle of continuity emphasizes that consistent and ongoing training is essential to maintain and improve performance. Interruptions or long breaks can lead to decreased fitness, reduced strength, and loss of coordination.

17. (b) I-4, II-3, III-2, IV-1

Principle of Overload : Training should be continuous (I-4)

Overload is about increasing intensity, not continuity.

Principle of Continuity : Training must be specific to the sport ((II-3)

Continuity focuses on consistency, not sport specificity.

Principle of Variance : Increasing training intensity gradually (III-2)

Variance prevents boredom, but overload increases intensity.

Principle of Specificity : Variety helps prevent boredom and injuries (IV-1)

Specificity focuses on sport-based training, not variety.

18. (c) Helmet

Helmets protect the head from injuries in contact sports like football, cycling, and cricket. Shin guards protect the lower legs. Gloves provide grip and protect hands. Knee pads cushion impacts to the knees.

19. (d) Reversibility

The Principle of Reversibility states that if training is stopped for an extended period, the fitness gains and skills achieved gradually decline. Regular exercise is necessary to maintain strength, endurance, flexibility, and overall performance.

20. (b) Three Periodisation is the planned variation of training intensity and volume over a cycle, often three weeks of hard training followed by a recovery week.

21. (b) Interval training Interval training alternates between high-intensity exercise and rest.

Continuous training (a) maintains a steady heart rate without rest.

Circuit training (c) includes different exercises but not fixed rest intervals.

Strength training (d) doesn't involve heart rate fluctuations like the graph.

22. (b) I-3, II-2, III-4, IV-1 Genetics : Determines an athlete's physical potential. (I-3)

Nutrition : Affects strength, recovery, and endurance. (II-2)

Motivation : Key to training consistency and competitive performance. (III-4)

Environment : Factors like altitude or temperature influence performance. (IV-1)

23. The objectives of training in sports are:

(i) Developing physical fitness, strength, and endurance.

(ii) Improving technical and tactical skills for better performance.

(iii) Enhancing psychological readiness, including focus, resilience, and stress management.

(iv) Building an all-rounded personality with discipline, confidence, and teamwork.

24. The important components for a successful training programme are:

1. **Define Goals** – Identify a primary goal and several minor goals to maintain focus.
2. **Match Sport to Goals and Abilities** – Align physical, emotional, and social needs with the chosen sport.
3. **Set Intermediate Goals** – Include general, seasonal, monthly, or weekly targets to track progress.
4. **Plan Workouts** – Structure training sessions into warm-up, skill practice, match-related practice, and cool-down phases.
5. **Get Professional Coach's Help** – Use a coach to correct form, analyze performance, and design effective programs.
6. **Join a Club or Organization** – Engage in group activities to enhance learning and motivation.
7. **Maintain Year-Round Fitness** – Utilize the off-season for rest and building strength and endurance without overtraining.
8. **Prevent Injuries** – Warm-up, stretch, and use proper techniques to avoid sport-specific injuries.
9. **Use Sport-Specific Training** – Focus on the skills, strength, and fitness required for the specific sport.
10. **Enjoy Training** – Stay motivated, avoid overtraining, and maintain a positive attitude for consistent improvement.

25. Principle of Overloading and Recovery in Sports Training:

(i) **Principle of Overload** – To improve performance, the training load must exceed the body's usual capacity. This involves gradually increasing the **type, intensity, or volume** of exercises so that the body adapts and becomes stronger. Overloading must be controlled to avoid fatigue or injury.

(ii) **Principle of Recovery** – Recovery is essential for the body to **regenerate and strengthen** after training. Proper rest periods between sessions prevent injuries, reduce fatigue, and ensure that fitness gains are effectively maintained.

26. Year-round fitness is important for an athlete because it ensures they remain in **peak physical condition**, reduces the risk of **injuries**, and supports **consistent performance**. By maintaining a balance of training, rest, and recovery throughout the year, athletes can sustain endurance, strength, and skill levels even during off-season periods.

27. In sports, **training** refers to the systematic and planned process of preparing an individual for an event, activity, or task to improve performance. **Sports training** is a long-term, structured program aimed at enhancing a sportsperson's physical, technical, psychological, and tactical abilities. It includes phases such as **preparation, competition, and transitional periods**. The ultimate goal is to develop an **all-rounded personality** and achieve **peak sports performance**, with coaches guiding the process to ensure effective skill and fitness development.

28. (i) Sports Training : It is a structured process aimed at improving an athlete's performance through physical, technical, tactical, and mental preparation.

(ii) (a) Conditioning Training : Improves strength, endurance, and flexibility, which are crucial for stamina and reducing injury risk.

(b) Training of Tactics : Enhances teamwork, decision-making and strategic game play to maximize match performance.

29. (i) **Principle of Specificity**: The sprinter's training focuses on exercises that target the specific demands of sprinting, such as speed, explosive power, and flexibility.

(ii) **Importance for Sprinters**: A sport-specific training plan ensures that every exercise directly enhances sprinting performance. It develops the necessary muscles, speed, and coordination while avoiding irrelevant training that could reduce efficiency or hinder race performance.

30. (i) **Principle of Reversibility**: This scenario highlights that fitness levels, including endurance, strength, and skill efficiency, decline when regular training is interrupted.

(ii) **Strategies to Maintain Fitness During Injury:**

(a) **Light or Alternative Training:** Perform low-impact activities such as swimming, cycling, or resistance band exercises to maintain cardiovascular fitness and muscle tone without stressing the injured area.

(b) **Physiotherapy and Mobility Exercises:** Incorporate guided physiotherapy, stretching, and mobility exercises to preserve joint flexibility, prevent stiffness, and support gradual recovery.

31. (i) Training of Technique: The scenario emphasizes training designed to develop cricket-specific skills such as batting, bowling, and fielding.

(ii) **Benefits of Sport-Specific Training:** Sport-specific training enhances performance by refining precise skills, movements, and tactics required for the sport. Unlike general physical fitness training, which improves overall health and endurance, this training directly targets abilities that impact success in cricket, leading to better game performance and efficiency.